

Extending Warehouse Order Processing System

In a real warehouse, orders arrive in **batches** — for example, all the orders received in a single day need to be processed together, and the warehouse manager wants a summary report at the end of the day, not just one order's result.

Extend your system to do the following:

1. **Collect multiple orders into a list.**
Ask the user how many orders they want to enter (e.g., n). Using a loop that runs n times, collect the same details as before (Customer ID, order value, number of items, delivery distance, premium member, priority order, fragile item) for each order — but instead of processing and printing immediately, store **each order as its own list** (e.g., [customer_id, order_value, num_items, distance_km, is_premium, is_priority, is_fragile]), and **append that list to a master list** called all_orders.
2. **Process every order in the batch.**
Loop through all_orders and, for each order, apply the same business rules as before (acceptance, free shipping, priority processing, special handling) and calculate its shipping cost. Print an individual summary for each order, just like the original system did.
3. **Generate an end-of-day report** using only the data now stored in all_orders (and any lists you build while processing them):
 - **Total number of orders received.**
 - **Number of orders ACCEPTED vs REJECTED.**
 - **Total revenue** from all accepted orders (sum of their order values).
 - **Total shipping cost collected** (only for accepted orders that did *not* qualify for free shipping).
 - **The single highest-value order** in the batch — print its Customer ID and order value.
 - **How many orders required special handling** (fragile or 10+ items).
 - **How many orders received priority processing.**